



Enhancing comprehensive diabetic care: A smartphone fundus camera with an offline AI-powered diabetic retinopathy screening solution for physicians

Kalpa Negiloni¹ · Divya Parthasarathy Rao² · Florian Mickael Savoy³ · Anand Sivaraman¹

Received: 27 May 2024 / Accepted: 10 July 2024 / Published online: 16 September 2024
© The Author(s) 2024

Dear Editor,

We congratulate the authors for their invaluable contribution in developing the diabetic retinopathy (DR) screening guidelines for Physicians in India: Position statement by the Research Society for the Study of Diabetes in India and the Vitreoretinal Society of India-2023 [1]. The study advocates for the implementation of fundus cameras alongside various strategies, such as employing DR screening AI algorithms, teleophthalmology consultations, and collaborating with ophthalmologists for effective DR screening crucial for early detection and timely intervention in DR.

We propose another India-centric device with an inbuilt AI for screening DR. The Medios AI - DR (CE MDR Class II, Europe and HSA, Singapore approved, Fig. 1) has demonstrated robust performance in identifying referable DR (moderate NPDR or worse, and/or CSDME). This solution integrated on a portable fundus camera (Remidio FOP NM-10, ~45degree, Fig. 1), can significantly enhance comprehensive diabetic care for physicians. Independent validation studies have found that the retinal image quality and DR detection are comparable to standard high-end desktop and other handheld cameras [2–4]. The significance

of Medios AI - DR lies in several key aspects: firstly, its seamless integration into a portable, smartphone-based non-mydratic fundus camera; secondly, its capability for offline operation, eliminating the need for internet connectivity to run the AI, generating a report in seconds; and thirdly, its prospective validation across various settings in large samples (~1000 patients), including diabetes care centers, communities, and hospitals alike in India and other ethnic populations. In validation studies conducted in India, the sensitivity ranged between 93% and 100%, and specificity ranged from 86.7% to 92.5% (Table 1). Notably, the sensitivity for detecting sight-threatening disease, which pose a risk of blindness, ranged between 98.7% and 100%, with no false negatives observed for cases involving active proliferative disease. Validation studies conducted in Yucatan peninsula, Dominican Republic, and Armenia underscored the generalizability of Medios AI - DR (offline) across different ethnic populations. Studies in India and Poland evaluated Medios AI - DR's effectiveness on images from existing screening systems like Topcon, showing promising device-agnostic capabilities (Table 1). Experts who have deployed and prospectively evaluated Medios AI - DR, emphasize its potential as a DR screening solution, particularly in developing countries requiring simple, affordable solutions with minimal infrastructure requirements [2, 5].

In summary, the adoption of AI-DR alongside fundus cameras while understanding the limitations as pointed out

Remidio Innovative Solutions Inc, USA, and Medios Technologies, Remidio Innovative Solutions Pte Ltd are wholly owned subsidiaries of Remidio Innovative Solutions, Pvt Ltd, India.

✉ Divya Parthasarathy Rao
drdivya@remidio.com

¹ AI&ML, Remidio Innovative Solutions Pvt Ltd, Bengaluru, India

² AI&ML, Remidio Innovative Solutions, Inc, Glen Allen, VA, USA

³ AI&ML, Medios Technologies, Remidio Innovative Solutions Pte Ltd, Singapore, Singapore



Fig. 1 Screening diabetic retinopathy using Medios AI - DR integrated to non-mydratric smartphone fundus camera (FOP NM-10) as a part of comprehensive diabetic care by physicians

Table 1 Results of the validation of Medios AI - DR in different ethnic populations and different screening settings

Setting	Study	Ethnicity	Sample (participants)	Study design	Sensitivity (95% CI)	Specificity (95% CI)
Community	Natarajan et al., JAMA Ophthalmol, 2019 ^a	Indian	231	Prospective	100.0% (78.2–100.0%)	88.4% (83.2–92.5%)
	Jain A et al. IJO, 2021 ^b	Indian	1294	Prospective	100% (94.72–100.00%)	89.55% (87.76–91.16%)
	Wroblewski et al., JDST, 2023 ^c	Mexican	248	Retrospective	Any DR: 94% (88–97%)	Any DR: 94% (88–98%)
	Polyclinics, Rao et al., IJO, 2024 ^d	Armenian	478	Retrospective	95.30% (91.9–98.7%)	83.89% (79.9–87.90%)
	Mobile primary health care, DR screening clinic, Kemp et al., BMJ Open Ophthal, 2023 ^e	Dominica	535	Prospective	81.4% (75.5–86.3%)	91.5% (87.9–94.3%)
Tertiary eye hospital	Rao et al., Clinical Ophthal, 2022 ^f	Indian	135	Retrospective (Topcon camera)	98.3% (96–100%)	83.7% (73%, 94%)
Diabetes Clinics	Gryzbowski et al., Opht Res, 2023 ^g	Polish	807	Retrospective (Topcon camera)	95% (91–98%)	80% (77–83%)
	Sosale et al. (BMJ Open Diab Res Care, 2020 ^h	Indian	900	Prospective	93% (91.3–94.7%)	92.5% (90.8–94.2%)
	Sosale et al., IJO, 2020 ⁱ	Indian	297	Prospective	98.84% (97.62–100%)	86.73% (82.87–90.59%)

^aNatarajan et al. JAMA Ophthalmology, 2019, PMID: 31,393,538

^bJain et al. Indian Journal of Ophthalmology, 2021, PMID: 34,708,760

^cWroblewski et al. Journal of Diabetes Science and Technology, 2023, PMID: 37,641,576

^dRao et al. Indian Journal of Ophthalmology, 2014 (Accepted for publication)

^eKemp et al. BMJ Open Ophthalmology, 2023, PMID: 38,135,351

^fRao et al. Clinical Ophthalmology, 2022, PMID: 36,003,071

^gGryzbowski et al. Ophthalmic Research, 2023, PMID: 37757777

^hSosale et al. BMJ Open Diabetes Research Care. 2020, PMID: 32,049,632

ⁱSosale et al. Indian Journal of Ophthalmology. 2020. PMID: 31,957,735

by the authors, and the careful selection of appropriate AI system by physicians, have immense potential to enhance patient care, improve screening accessibility and efficiency, crucial for early detection and intervention in DR.

Author contribution Kalpa Negiloni and Divya Parthasarathy Rao participated in conceptualization and literature review. Kalpa Negiloni participated in writing the original draft. Divya Parthasarathy Rao, Florian Mickael Savoy, and Anand Sivaraman participated in editing and reviewing the manuscript.

Declarations

Competing interests All the authors are employees of Remidio Innovative Solutions. FMS has patents and financial interests in Remidio Innovative Solutions Pvt Ltd (ESOP and Stock). AS has patents, stock, and leadership role.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Agarwal M, Rani PK, Raman R, Narayanan R, Virmani A, Rajalakshmi R, Chandrashekhar S, Makkar BM, Agarwal S, Palanivelu MS, Srinivasa MN. Diabetic retinopathy screening guidelines for Physicians in India: position statement by the Research Society for the Study of Diabetes in India (RSSDI) and the Vitreoretinal Society of India (VRSI)-2023. *Int J Diabetes Dev Countries*. 2024;20:1–8.
2. Rajalakshmi R, Subashini R, Anjana RM, Mohan V. Automated diabetic retinopathy detection in smartphone-based fundus photography using artificial intelligence. *Eye*. 2018;32(6):1138–44.
3. Sengupta S, Sindal MD, Baskaran P, Pan U, Venkatesh R. Sensitivity and specificity of smartphone-based retinal imaging for diabetic retinopathy: a comparative study. *Ophthalmol Retina*. 2019;3(2):146–53.
4. Das S, Kuht HJ, De Silva I, Deol SS, Osman L, Burns J, Sarvananthan N, Sarodia U, Kapoor B, Islam T, Sampath R, Poyser A, Konidaris V, Anzidei R, Proudlock FA, Thomas MG. Feasibility and clinical utility of handheld fundus cameras for retinal imaging. *Eye (Lond)*. 2023;37(2):274–9.
5. Grzybowski A, Singhanetr P, Nanegrungsunk O, Ruamviboonsuk P. Artificial intelligence for diabetic retinopathy screening using color retinal photographs: from development to deployment. *Ophthalmol Therapy*. 2023;12(3):1419–37.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.